NEW UZBEKISTAN UNIVERSITY

# Week 3
# Data Cleaning and Quality



Decisions that change what data can tell us

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# Week 3 learning objectives

CLO1: explain preprocessing. CLO2: clean data appropriately.

| | |
|---|---|
| **Quality dimensions** | Explain six quality dimensions and choose checks for a stated question. |
| **Cleaning decisions** | Justify format fixes and duplicate rules. Distinguish unusual values from errors. |
| **Missingness** | Explain MCAR, MAR, and MNAR using examples and state the assumptions. |
| **Audit trail** | Record each decision, count its effect, and check the cleaned data with assertions. |

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# The connection to Week 2

A summary depends on which values enter the calculation

## Last week

We described centre, spread, and shape.

We reported the unit and the number of valid observations.

## This week

We inspect the values behind those summaries.

We ask whether cleaning changes the population or the conclusion.

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**A correct formula can still summarize the wrong data.**

# A registry export arrives

Illustrative records. One row should describe one student in the current registry.

| ID | Programme | Age | Score | Updated |
|---|---|---|---|---|
| A01 | Biology | 19 | 72 | 2026-09-01 |
| A02 | biology | 20 | 67% | 2026-09-01 |
| A02 | Biology | 20 | 69 | 2026-09-03 |
| A03 | Economics | 220 | blank | 2026-09-01 |
| A04 | Economics | 47 | 88 | 2026-09-01 |

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**Discussion: What needs investigation before you calculate an average?**

# One repeated record changes an average

Illustrative scores. The second A02 row repeats the same measurement.

| Record | Score |
| --- | --- |
| A01 | 40 |
| A02 | 80 |
| A02, repeated copy | 80 |

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**Mean of rows = 66.7**

**Mean of students = 60.0**

**The repeated record gives one student twice the weight.**

# Quality is fitness for a purpose

A cell holds one value. This student table contains one missing phone-number cell.

| Student ID | Programme | Phone |
|---|---|---|
| 101 | AI | 555-0101 |
| 102 | AI | **(missing)** |
| 103 | Robotics | 555-0103 |

Count students by programme
AI: 2. Robotics: 1. The blank phone does not affect these counts.

Make urgent phone calls
Student 102 cannot be called using this table. The blank matters here.

**Data quality depends on whether the data support the intended task.**

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# Six dimensions of data quality

Required records, valid values, and agreement between fields

| | |
|---|---|
| **Completeness** | Check missing cells and absent records. Example: an export contains 95 of 100 expected students. |
| **Validity** | Do values follow a documented type, category, or range? Example: score from 0 to 100. |
| **Consistency** | Do related values agree? Example: an end date should not precede its start date. |

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# Six dimensions of data quality

Second: identity, reality, and time

| | |
|---|---|
| **Uniqueness** | A key identifies a row. A current student registry expects one row per student ID. |
| **Accuracy** | Does the recorded value match the true value? Verification may require the original record. |
| **Timeliness** | Is the information recent enough for this decision? Check the event date and snapshot date. |

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# A valid value can still be wrong

Valid means it follows a rule. Accurate means it matches the correct value for that student.

| Stored score | 0–100 range check | Verified marked exam: 76 |
|---|---|---|
| 760 | Fails | Outside the permitted score range. |
| 67 | Passes | Wrong: someone reversed the digits of 76. |
| 76 | Passes | Matches the verified exam record. |

**Why? A range check accepts both 67 and 76. It cannot tell which score the student earned.**

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# An inventory records the starting condition

An inventory is a short summary before cleaning. It reveals problems and lets us explain later changes.

| Check | Example before cleaning | Why record it? |
|---|---|---|
| Rows and IDs | 12 rows, 10 distinct student IDs | Repeated IDs need investigation. |
| Missing values | 2 of 12 phone cells are missing | Compare missingness before and after. |
| Types and ranges | Age includes 220 and text “21” | Check the age error and the stored type. |

**After cleaning: 12 rows become 10 only if we can explain the two removed rows.**

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# The data dictionary defines the checks

Example dictionary for an illustrative registry

| Field | Meaning | Documented rule |
| --- | --- | --- |
| student_id | Identifier | Text. Required and unique in current registry. |
| entry_score | Admission result | Numeric points from 0 to 100. |
| programme | Degree programme | One of the approved programme names. |
| updated_at | Record revision time | Timestamp with documented timezone. |

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# Different labels can describe one category

Software counts different text labels as separate groups. This can make Biology look like three programmes.

| Raw programme label | Students | Meaning confirmed by dictionary |
|---|---|---|
| “ biology ” | 4 | Biology. Remove outside spaces. |
| “Biology” | 3 | Biology. Keep the approved label. |
| “BIO” | 2 | Biology. Apply the approved abbreviation map. |
| “Bioengineering” | 1 | A different programme. Keep it separate. |

**After standardising: Biology = 4 + 3 + 2 = 9 students. Bioengineering = 1.**

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# Stored types and units affect calculations

Type: text or number. Unit: what the number measures. Representation: how we write it. We need numeric values to calculate a mean, and matching units to compare them.

| Raw text | Meaning | Correct stored value |
|---|---|---|
| “67%” | Percentage | Number 0.67 as a proportion |
| “0.67” | Proportion from 0 to 1 | Number 0.67, equivalent to 67% |
| “00124” | Student identifier | Text “00124” keeps leading zeros |
| “1,200” | Separator convention unclear | Confirm whether it means 1200 or 1.2 |

**The data dictionary explains meanings and units. Conversion follows that definition.**

# A date parser needs the source format

A parser is software that reads text using a stated format. It turns date text into a date so we can sort events or calculate durations. What does “03/04/2026” mean?

| Format supplied to the parser | Date it reads | Clear output |
|---|---|---|
| Day / month / year | 3 April 2026 | 2026-04-03 |
| Month / day / year | 4 March 2026 | 2026-03-04 |

A wrong choice moves the event by 30 days. Both results are possible dates.

**Check the source’s date convention first. Keep unresolved dates flagged until confirmed.**

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# Numeric conversion and new missing values

A numeric parser turns text such as “72” into the number 72 for calculations.
In this example, failed conversions become missing-value markers.

| Raw score text | Numeric result | What happened? |
|---|---|---|
| “72” | 72 | Conversion succeeds. |
| (blank) | missing | The raw file already lacked a value. |
| “seventy” | missing | New missing value: this parser cannot read words. |

**Missing cells increase from 1 to 2. Keep the raw text and investigate the failed conversion.**

# Three meanings of repeated records

The unit of observation means what one row represents, such as one student or one weekly result.

| Case | Two example records | Appropriate response |
|---|---|---|
| Exact copy | ID 101, score 70<br>ID 101, score 70 | For a current registry: confirm a copied row, then keep one. |
| Conflicting ID | ID 102, score 60<br>ID 102, score 80 | Verify the correct score or version. Keep the conflict flagged meanwhile. |
| Valid repetition | ID 103, week 1, score 70<br>ID 103, week 2, score 80 | For weekly results: keep both. The key includes ID and week. |

**A repeated student ID is a problem only when the intended row definition requires it to be unique.**

# A conflicting ID needs a rule

Illustrative revisions from a system that explicitly tracks corrections

| ID | Score | updated_at | Status |
|---|---|---|---|
| A02 | 67 | 2026-09-01 09:00 | Original |
| A02 | 69 | 2026-09-03 11:00 | Verified correction |

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**Here, the source confirms that the verified correction supersedes the original.**

**Keep the original in the audit history.**

# When “keep the latest” is unsafe

A timestamp only helps when its meaning is clear

| | |
|---|---|
| **Registration date** | The student registered later. That does not prove the score is a correction. |
| **Conflicting sources** | A recent spreadsheet may be less authoritative than the signed record. |
| **Unresolved conflict** | Retain both versions for review and flag the field or record as unresolved. |

**The cleaning rule should name the evidence that makes one record authoritative.**

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# Discussion: which repeated rows should stay?

These three small tables have different row meanings. For each pair, choose a key and justify an action.

| Table | Row | ID / invoice | Other recorded fields |
|---|---|---|---|
| Enrolments | A | Student 101 | Course DS |
| Enrolments | B | Student 101 | Course CV |
| Purchases | C | Invoice 501 | Same purchase, total 100 USD |
| Purchases | D | Invoice 501 | Same purchase, total 100 USD |
| Current registry | E | Student 102 | Birth date 2006-02-10 |
| Current registry | F | Student 102 | Birth date 2006-10-02 |

**Which pair stays, which confirmed copy can go, and which needs source evidence? Why?**

# Reasoning about repeated rows

A useful answer includes the unit and the evidence

| | |
|---|---|
| **Two courses** | Keep both registrations. Student ID alone is not the registration key. |
| **Duplicate load** | Keep one purchase after verifying that both rows represent the same invoice event. |
| **Birth-date conflict** | Check an authoritative source. An arbitrary first or last row has no evidential advantage. |

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# Question 1: is the reported mean justified?

Five minutes. One current row per student. BIO and “ biology ” mean Biology.
The source has not confirmed which score for 00102 is correct. Scores use 0–100 points.

| Row | Student ID | Programme | Score | Recorded date |
|---|---|---|---|---|
| A | 00101 | Biology | 70 | 2026-09-01 |
| B | 00101 | BIO | 70 | 2026-09-01 |
| C | 00102 | biology | 60 | 2026-09-01 |
| D | 00102 | Biology | 80 | 2026-09-03 |
| E | 00103 | Bioengineering | 90 | 2026-09-01 |

B is a confirmed copy of A. An analyst keeps the latest row per ID and reports Biology mean = 75.
1. Is 75 justified? Explain.  2. Which changes are safe, and which need evidence?
3. Using the two candidate scores for 00102, calculate both possible Biology means.

**Email during class to m.sabokrou@newuu.uz. Subject: STUDENTID_W3_Q1**
**Body: class group, reasoning in your own words, and calculations. About 80–120 words.**

# Missing values

A blank can change both the denominator and the conclusion

## Who did not appear in this number?

The reason for missingness matters as much as the amount.

# Zero, missing and not applicable differ

Example: “How many hours did you use the campus laboratory this week?”
The question applies only to students enrolled in the campus lab module.

| Student | Recorded hours | Meaning |
|---|---|---|
| 101: enrolled | 0 | Tracked for the week. No lab use occurred. |
| 102: enrolled | (missing) | Tracking failed. Actual hours could be 0, 2 or another value. |
| 103: not enrolled | Not applicable | Outside this question’s target group. |

**Changing student 102’s blank to 0 claims “no use”, although we do not know the hours.**

# Completeness needs a denominator

Course-satisfaction survey: 1 = very dissatisfied, 5 = very satisfied. We invite 20 students.

| Quantity | Value |
|---|---|
| Students invited | 20 |
| Students who answered | 12 |
| Missing answers | 8 |
| Completeness of answers | 12 ÷ 20 = 60% |

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**Completeness tells us how much is missing.**
**A mechanism describes why answers are missing and who is less likely to reply.**

# MCAR: loss unrelated to students or answers

MCAR means missing completely at random. Example: random technical loss of survey forms.

| Assumed process | What it means |
| --- | --- |
| 20 ratings collected, 8 randomly lost | 12 remain. Completeness = 12 ÷ 20 = 60%. |
| Every form has the same loss chance | Loss ignores class group and satisfaction rating. |
| We analyse the 12 remaining ratings | No systematic selection toward higher or lower ratings. |

**Completeness: 60% of answers remain. MCAR: the loss process is random. Fewer answers mean greater uncertainty in the estimate.**

# MAR: reply chance depends on known class

MAR = missing at random given observed information. Class is known for everyone. Missing value: course satisfaction. The table gives assumed reply chances.

| Known class | Satisfied: reply chance | Dissatisfied: reply chance |
|---|---|---|
| Day | 90% | 90% |
| Evening | 40% | 40% |

Suppose evening students have less time to reply. Within either class, satisfied and dissatisfied students have the same chance of answering.

**The missing rating does not change reply chance after we account for class.**

# MNAR: satisfaction still affects replies

MNAR means missing not at random. This table assumes a different evening reply process. We still know every student's class, but nonresponders' ratings remain unknown.

| Known class | Satisfied: reply chance | Dissatisfied: reply chance |
|---|---|---|
| Day | 90% | 90% |
| Evening | 40% | 10% |

Suppose dissatisfied evening students avoid the survey. Even within evening class, their chance of replying is lower. Replies can make satisfaction look higher.

**MAR on the previous slide: 40% = 40%. MNAR here: 40% ≠ 10% within evening class.**

# MCAR, MAR and MNAR describe missingness

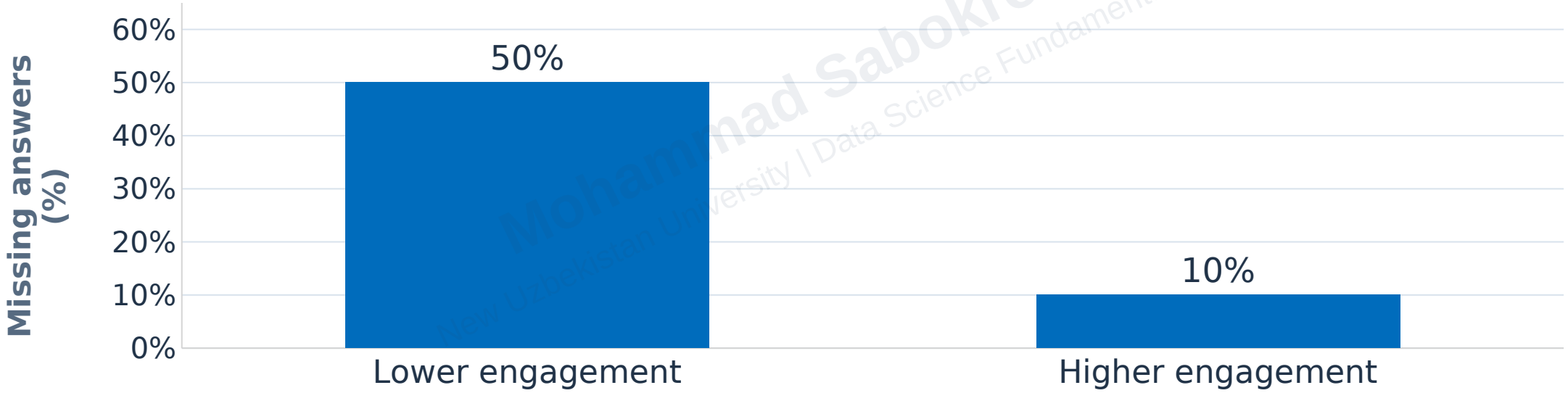
A mechanism describes the process that makes a value missing. These names classify assumptions.
Same survey throughout: known class, potentially missing course-satisfaction rating.

| Assumption | Example process | What changes reply chance? |
|---|---|---|
| MCAR | Random loss of forms | Neither class nor satisfaction. |
| MAR | Evening students have less time | Class matters. Within class, satisfaction does not. |
| MNAR | Dissatisfied evening students avoid replying | Satisfaction matters even within the same class. |

**These are assumptions about missingness, not measured quality scores. Observed answers alone generally cannot tell MAR from MNAR.**

# Different response rates need an explanation

Course-satisfaction survey: 20 students per group. Here, engagement means attendance. Lower group: 5 of 10 classes attended. Higher group: 9 of 10 classes attended.



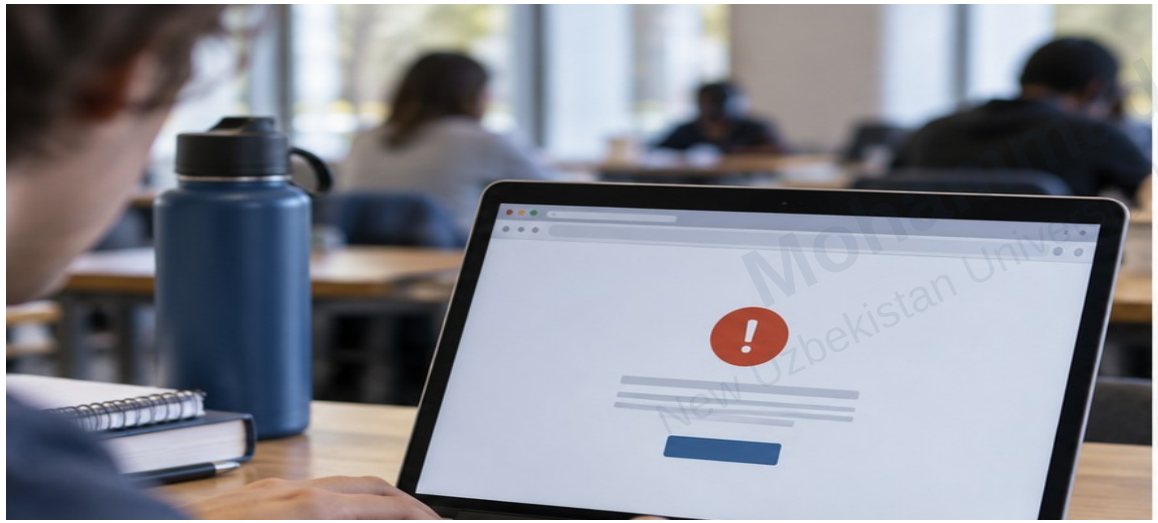

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**Missing ratings: lower group 10/20 = 50%, higher group 2/20 = 10%. Possible causes: missed reminders or dissatisfaction. The rates do not prove MAR.**

# The same missing rate can hide different bias

Each illustrative programme invites 20 students and has 12 satisfaction ratings.
Missing: 8/20 = 40% in both programmes.

## A: random technical loss



Eight randomly chosen forms are lost.
No systematic preference for satisfied students.

## B: dissatisfied students skip



The eight most dissatisfied students skip.
The replies make the programme look happier.

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# Deleting incomplete rows changes who remains

Illustrative class. A required survey field has different response rates.

| Group | Original students | Complete rows | Retained |
|---|---|---|---|
| Day group | 80 | 72 | 90% |
| Evening group | 20 | 8 | 40% |
| Total | 100 | 80 | 80% |

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**Evening students fall from 20% of the class to 10% of the complete cases.**

# Mean imputation reduces the visible spread

Imputation fills missing values. Here, three blanks receive the observed mean of 3.

| Treatment | Displayed values | Mean | Sample SD |
|---|---|---|---|
| Observed values only | 2, 4 | 3 | 1.41 |
| Fill blanks with 3 | 2, 4, 3, 3, 3 | 3 | 0.71 |

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**The mean is unchanged. The apparent variation is smaller.**

**Five filled cells still contain only two observed answers.**

# Sensitivity analysis makes assumptions visible

20 invited students. 12 responses have mean 4.0 on a 1–5 scale.

**Observed total = 12 × 4 = 48**

Let m be the average answer of the 8 missing students.

**Mean for all 20 = (48 + 8 × m) ÷ 20**

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**We vary m to see whether the conclusion depends on answers we do not know.**

# Different assumptions, different conclusions

The target is an overall mean of at least 3.5

| Assumed missing mean m | Overall mean | Meets 3.5? |
|---|---|---|
| 1.0 | 2.8 | No |
| 3.0 | 3.6 | Yes |
| 4.0 | 4.0 | Yes |
| 5.0 | 4.4 | Yes |

**With scores limited to 1–5, the overall mean can range from 2.8 to 4.4.**

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# Recover the missing value from its source

A registry export has no score for student 201. The verified original exam record gives 78.

| Record | Raw export | After source verification |
| --- | --- | --- |
| Student 201 score | missing | 78 points |

**We recover an existing measurement. Match the student and exam before copying the value.**

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# Keep the missing value and report the limit

Scores are 70, 80 and missing for three students. We want to describe what we actually observed.

| Quantity | Result |
| --- | --- |
| Mean of observed scores | (70 + 80) ÷ 2 = 75 |
| Coverage | 2 observed scores out of 3 students |

**We can report “observed mean = 75, n = 2 of 3”. We cannot claim the all-student mean is 75.**

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# Exclude rows for a stated analysis

The same scores are 70, 80 and missing. An analyst keeps only rows with a recorded score.

| Student | Score | Included in this analysis? |
|---|---|---|
| 201 | 70 | Yes |
| 202 | 80 | Yes |
| 203 | missing | No |

**The result describes two included students. Excluding a row can change who the analysis represents.**

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# Impute an assumed replacement value

Imputation fills a blank with an estimated or assumed value. Here, the analyst fills it with the observed mean.

| Student | Observed score | Stored score after filling |
|---|---|---|
| 201 | 70 | 70 |
| 202 | 80 | 80 |
| 203 | missing | 75, imputed |

**The mean stays 75, but student 203’s score is still unknown. Mark the filled value as imputed.**

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# Summary: four responses to missing data

Lab 3: diagnose missingness and keep missing values unfilled.

| Option | When it can help | What to report |
| --- | --- | --- |
| Recover the value | A reliable source or follow-up exists. | Source and recovery process. |
| Leave it missing | Analysis can use other observed fields. | Valid n and missingness pattern. |
| Exclude observations | The target and missingness assumptions justify it. | Retention and group differences. |
| Impute values | A justified model and uncertainty method exist. | Assumptions and sensitivity. |

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# Unusual values need an explanation

Illustrative ages in an undergraduate registry

| Age | Interpretation | Initial response |
|---|---|---|
| 220 | Impossible human age. | Check source. Mark invalid if unresolved. |
| 47 | Unusual but plausible student. | Keep and verify if necessary. |
| 21 | Plausible, but may still be mistyped. | Accuracy needs source evidence. |

**Distance from the average does not determine whether a record is wrong.**

# The IQR rule flags values for review

Q1 (25th percentile) = 19. Q3 (75th percentile) = 22. Ages are in years.

## IQR = Q3 − Q1 = 3 years

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### Lower fence

$Q1 - 1.5 \times IQR$
$19 - 1.5 \times 3 = 14.5$

### Upper fence

$Q3 + 1.5 \times IQR$
$22 + 1.5 \times 3 = 26.5$

**Both ages 47 and 220 exceed the upper fence. Their explanations are very different.**

# A threshold can remove a subpopulation

Illustrative cohort. All 5 students above age 35 study part-time.

| Population | Full-time | Part-time | Total |
|---|---|---|---|
| Original cohort | 80 | 20 | 100 |
| After deleting age > 35 | 80 | 15 | 95 |

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**Part-time share changes from 20.0% to 15.8%.**

**The cleaned table now represents a different mix of students.**

# Influence can be reported without deletion

Withdrawal means leaving the course. The goal is to describe all 100 enrolled students. Five students are older than 35. Their ages are valid, and four of them withdrew.

| Students counted | Withdrawals | Students | Withdrawal rate |
|---|---|---|---|
| All students | 10 | 100 | 10 ÷ 100 = 10.0% |
| Older than 35 | 4 | 5 | 4 ÷ 5 = 80.0% |
| After excluding the older group | 10 − 4 = 6 | 100 − 5 = 95 | 6 ÷ 95 = 6.3% |

**Deleting the five older students removes four withdrawals: 10% becomes 6.3%. This changes the population. Keep the all-student result and explain the comparison.**

# A defensible response to an outlier

The evidence determines the action

| | |
|---|---|
| **Verified entry error** | Correct it from the source. Preserve the raw value and record the reason. |
| **Invalid, unresolved value** | Flag the value and retain usable fields where appropriate. Do not guess a replacement. |
| **Valid, influential value** | Keep it. Use suitable summaries and report sensitivity when the result changes. |

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# Question 2: does this survey meet the target?

Five minutes. Satisfaction uses 1–5. The course target is an all-student mean of at least 3.5. An analyst pools the replies and claims: “Mean = 3.8, so the course meets the target.”

| Known class | All students | Replies | Missing | Reply mean |
|---|---|---|---|---|
| Day | 80 | 72 | 8 | 4 |
| Evening | 20 | 8 | 12 | 2 |

1. Assume replies represent each class. Estimate the mean using the full class sizes.
2. Do different reply rates prove MAR? Explain what must hold within each class.
3. Now assume the 12 missing evening ratings average 1, and all day students average 4. Recalculate the all-student mean. What happens to the target claim?

**Email during class to m.sabokrou@newuu.uz. Subject: STUDENTID_W3_Q2**
**Body: class group, reasoning in your own words, and calculations. About 80–120 words.**

# Examples behind the cleaning log

Issue = what needs attention. Affected = how many rows or values.
Action = what you changed or kept. Reason = the evidence for that decision.

| Before | Action and result | Reason |
|---|---|---|
| Four rows: “ biology ” | Trim spaces and use “Biology” | Dictionary confirms the same programme. |
| One confirmed copied row | Keep one copy, remove the extra row | Both rows describe the same record. |
| One student: age 220 | Keep student, flag age as missing | Invalid age, correct age unknown. |
| One student: age 47 | Keep the row and age | Unusual but plausible for this population. |

**Log the units: 4 rows edited; 1 copy removed; 1 value flagged; 1 record retained.**
**Next: the summary cleaning log.**

# The cleaning log explains every decision

An illustrative audit record

| Issue | Affected | Action | Reason |
|---|---|---|---|
| Whitespace in programme | 4 rows | Trim spaces | Same approved category. |
| Verified duplicate load | 1 extra row | Retain one | Same record and event. |
| Age 220 | 1 value | Flag as missing | Invalid. No verified correction. |
| Age 47 | 1 record | Keep | Plausible member of target cohort. |

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# Assertions and accuracy checks

Assertions stop on a failed rule. Accuracy checks compare with a trusted source.

| Rule or source check | Example check result | Next action |
|---|---|---|
| Current ID must be unique | 2 IDs repeat | Inspect conflicts before analysis. |
| Score must lie from 0 to 100 | 1 score equals 760 | Review and resolve or flag. |
| Programme must be approved | 1 label is unknown | Check dictionary or source. |
| Compare with the original record | Not yet checked | Verify or report the limitation. |

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# Row accounting separates different changes

Illustrative registry export with 12 rows

| Step | Rows | What changed |
|---|---|---|
| Raw export | 12 | Starting point. |
| Remove 2 confirmed exact copies | 10 | Two extra copies removed. |
| Resolve 1 verified earlier version | 9 | One current row per student. |
| Flag 1 invalid age as missing | 9 | One age changes. Student stays. |

**A cleaning log should explain both row removals and value changes.**

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# Case discussion: a small registry

One row per current student. Scores use a documented 0–100 scale.

| ID | Programme | Age | Score | Record status |
|---|---|---|---|---|
| B01 | Biology | 19 | 70 | Current |
| B01 | Biology | 19 | 70 | Confirmed extra copy |
| B02 | biology | 220 | 80% | Current |
| B03 | Economics | 47 | blank | Current |

**Choose actions. Then report the mean score and its limitation.**

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# Case discussion: a defensible solution

Three students remain after removing the confirmed extra copy

| ID | Programme | Age | Score |
|---|---|---|---|
| B01 | Biology | 19 | 70 |
| B02 | Biology | missing, flagged | 80 |
| B03 | Economics | 47 | missing |

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**Observed score mean = (70 + 80) ÷ 2 = 75**

**This is the mean for 2 observed scores among 3 students.**

# A result needs an interpretation and a limit

Example reporting paragraph for the small registry

“After removing one confirmed duplicate, the registry contains three students. Two have recorded entry scores, with a mean of 75 out of 100. One score remains missing. We corrected documented formatting and flagged an invalid age. We cannot conclude that the mean score of all three students is 75.”

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**The paragraph identifies the population, denominator, action, and uncertainty.**

# What should survive every cleaning step?

A short explanation is part of the analysis

| | |
|---|---|
| **Meaning** | The variable, unit, population, and key still have clear definitions. |
| **Evidence** | Each correction or exclusion has a reason that another analyst can inspect. |
| **Uncertainty** | Missingness, unresolved conflicts, and the effect of assumptions remain visible. |

**A defensible cleaning decision can be explained in an ordinary sentence.**

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# Questions to take away

Explain your reasoning before looking back at the worked examples

**1** A value is four standard deviations from the mean. What evidence would justify removal?

**2** Two groups have the same response rate. What could still bias their comparison?

**3** All checks pass. Which kind of quality problem may still remain?

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# Lab 3 and the next lecture

Chapter 4: Data Quality and Cleaning

## Lab 3: registry cleaning

Submit the notebook, cleaning log, and cleaned CSV.

Inventory issues, justify each fix, and run assertions.

Keep missing values unfilled.

## Week 4

Transformation, features, and leakage.

How representation and the order of preprocessing affect an analysis.

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**Reading: Chapter 4. Record the evidence and limitation for each cleaning decision.**

# References

Course material and technical documentation

Course plan and Fundamentals of Data Science, Vol. 1, Chapter 4

Seaman et al. (2013): What Is Meant by “Missing at Random”?

NIST/SEMATECH handbook: Outliers (§7.1.6), Box Plot (§1.3.3.7)

pandas: to_numeric and DataFrame.drop_duplicates

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**All short classroom datasets in this lecture are illustrative.**